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Can I Have a Causal Graph, Please?

Welcome to *Chapter 12*.

We will start our journey to the land of **causal discovery** here.

Most people who hear about causal discovery for the first time are truly fascinated by this topic. Other people are skeptical, yet almost no one is indifferent.

Perhaps this tendency toward stronger reactions reveals something fundamental about our own human nature. In the first chapter, we briefly discussed Alison Gopnik's research showing that even infants and small babies run experiments in order to efficiently construct world models.

In this chapter, we'll discuss three sources of causal knowledge and we'll think together about the relative advantages and disadvantages of using them.

By the end of this chapter, you'll have a clear idea about different sources of causal knowledge and will be able to discuss their core strengths and weaknesses.

In this chapter, we will cover the following:

- Sources of causal knowledge
- Scientific experiments and simulations as a source of causal knowledge
- Personal observations and domain knowledge
- Causal discovery and structure learning

Sources of causal knowledge

Malika, Illya, and Lian are 11-month-old infants playing together. Malika passes a toy car to Illya, who had dropped it accidentally. Lian is trying to grab a new teddy bear that her sister put next to the chair. If you observe Malika, Illya, and Lian, you will quickly notice that they are very motivated to explore the environment around them.

This is a good sign – one of the indicators of a healthy developmental trajectory.

There's a catch though.

How do they choose *what* to explore?

You and I, oversaturated

It's estimated that around 11 million bits of information are sent from sensory inputs to the brain every second. In adults, only about 60 bits per second can be consciously processed (Martín, 2009).

It's difficult to assess how much of the 11 million bits can be processed unconsciously. It is also difficult to translate these numbers directly into babies' processing capabilities, but it's safe to say that infants are not able to process all the information available in the environment most of the time.

One of the mechanisms that helps humans and other animals to free up some of their processing capacity is called **habituation**. Habituation is a process in which an organism's response to repeated or prolonged presentations of a stimulus decreases. In other words, we stop paying attention to things that happen around us repeatedly over and over again in a similar way.

Have you ever moved to a new house or apartment with a loud street behind the windows? If you had this experience, it's likely that it was hard not to notice the noise in the beginning.

There's a chance it was pretty annoying to you. You might have even said to yourself or your family or friends "Gosh, how loud this is!"

It is likely that in a month or so, you stopped noticing the noise. The noise became a part of the background; it became essentially *invisible*.

This is habituation at work.

Habituation appears very early in development and can be observed even in later pre-natal stages (e.g., Muenssinger et al., 2013).

The power of a surprise

Developmental psychologists took advantage of the habituation mechanism and created research paradigms such as the **violation of expectation (VoE)**, which allow them to study how little babies learn. Researchers have shown that surprising (*unhabituated*) events can trigger multiple specific reactions such as alterations in facial expressions, pupil dilation, changes in cerebral blood flow, changes in brain electrical activity, and alterations in looking time.

Aimee Stahl and Lisa Feigenson from Johns Hopkins University have shown groups of 11-month-old infants regular toys and toys that (seemingly) violated the laws of physics. The *altered* toys went through the walls, flowed in the air, and suddenly changed location.

Infants spent more time exploring the altered toys than the regular ones. They also engaged with them differently.

The analyses have shown that babies learned about objects with unexpected behavior better, explored them more, and tested relevant hypotheses for these objects' behavior (Stahl & Feigenson, 2015).

This and other experiments (see Gopnik, 2009) suggest that even very small babies systematically choose actions that help them falsify hypotheses and build relevant causal world models. This interventional active learning approach is essential in achieving this goal.

Do you remember Malika, Illya, and Lian from the beginning of this chapter? Malika passed a toy car to Illya, who dropped it accidentally. It turns out that it's much more likely that a baby will pass a dropped object to a person who dropped it accidentally than to someone who dropped it or threw it intentionally.

World models are critical for efficient navigation in physical and social worlds, and we do not stop building them as grown-ups.

Scientific insights

Human babies have a lot to learn during their development – the laws of physics, language, and social norms. After learning the basics, most children start school. School provides most of us with a canon of current knowledge about a broad variety of topics.

To many people, learning this canon is sufficient. Others, unsettled by the answers to the questions they have or motivated by other factors, decide to dig deeper and look for the answers themselves. Some of these people become scientists.

In this section, we'll take a look at causal knowledge through the lens of scientific methods. We'll introduce some core scientific methodologies and touch upon their benefits and limitations.

The logic of science

Karl Popper, one of the 20th century's most famous philosophers of science, used to say that "*what is true in logic is true in scientific method*" (Popper, 1971). We briefly mentioned Popper's framework in *Chapter 7* and we'll refresh it now.

Popper's narrative is as follows: saying that all *Xs* are *x* is usually impossible because we usually cannot observe all *Xs* (and if in certain cases we can observe all *Xs*, we don't need science for these cases, as we can observe all the outcomes).

Although we cannot observe all *Xs* and hence we cannot definitely confirm any particular hypothesis about *Xs*, we can still learn something useful.

"All *Xs* are *x*" is logically equivalent to "no *X* is *not x*." Although we cannot unambiguously prove that all *Xs* are *x*, finding even a single case where *X* is *not x* disproves the hypothesis that all *Xs* are *x*. Disproving a hypothesis in this way is called **falsification**.

A **falsifiable hypothesis** is a hypothesis for which we can define an empirical test that can falsify this hypothesis. *All pigeons are white* is a falsifiable hypothesis that can be falsified by finding at least one non-white pigeon.

On the other hand, the existence of an intelligent supernatural being is usually not falsifiable unless we set a hypothesis in a very specific way, for instance, *the existence of a supernatural being guarantees that no person ever dies before their 18th birthday*.

It seems to me that not many people who advocate for the existence of supernatural beings are interested in formulating hypotheses such as this. Other, more general or more vague hypotheses in similar contexts usually turn out to be unfalsifiable.

Hypotheses are a species

In the Popperian framework, hypotheses come from theories (or tentative theories). Once generated, hypotheses should be scrutinized, i.e., put to the empirical test, which can falsify them.

If a hypothesis generated from a theory is falsified, the theory in its original shape is falsified and should be retracted. Every theory can generate many different hypotheses, yet a theory that survives the largest number of tests is *not necessarily more true*. It's more *fit*, in a sense, analogous to *fitness* in biological evolution.

For instance, a theory might *fit* the current data very well. The reason for that might be that we simply do not have advanced enough tooling to collect the data that *could falsify* this theory and so the theory's *fitness* does not automatically translate to its *truthfulness*. Some of these views have been since criticized and not all of them are broadly accepted today (see the callout box).

Popper and other philosophers

We present a very brief and synthetic overview of the Popperian framework here, which is certainly simplified and not complete. Although Popper's ideas are still a vital part of modern scientific culture, we should be aware that some of them have been criticized.

For instance, Thomas Kuhn has proposed an alternative model of how science develops over time (Kuhn, 1962), and other philosophers such as Imre Lakatos or Paul Feyerabend criticized Popper's views on the demarcation between science and pseudo-science and his criteria of theory rejection.

One of the (perhaps surprising) consequences of the consistent Popperism is the rejection of Darwin's theory of evolution as non-scientific (Popper himself called it *an interesting metaphysics project*; Rosenberg & McIntyre, 2020).

French physicist Pierre Duhem and American logician Willard Van Orman Quine argued independently that hypotheses cannot be empirically falsified in isolation because an empirical test needs auxiliary assumptions. Popper's theory has been also criticized as overly romantic and non-realistic (Rosenberg & McIntyre, 2020).

One logic, many ways

Scientific inquiry can take different forms depending on the object of interest of a given field.

Physicists usually design their experiments differently from psychologists. Although virtually all scientists follow (or at least do their best to follow) a set of similar basic principles, they might *do* very different things *in practice*.

For instance, physicists are interested in formulating hypotheses regarding the universal basic laws that rule the physical world. The objects they are interested in can be often described by a finite number of well-defined physical observables.

On the other hand, psychologists try to capture universal laws governing human and other animals' cognition and behavior.

The number of ways in which two individual chimpanzees or humans can differ is even difficult to quantify. Animals can be viewed as complex systems, where multiple different factors interact with each other and with the environment in a virtually infinite dance.

These differences are reflected in the choice of scientific tools that scientists in these fields make.

Controlled experiments

Physicists, chemists, and other representatives of the so-called *hard sciences* often work with controlled experiments. The *control* aspect means that the experimental environment is carefully controlled in order to minimize any interference from the outside world or – in other words – to keep the context variables constant. An extreme example of such control is projects such as the *Large Hadron Collider*

– a sophisticated experimental system consisting of a 27-kilometer-long ring of superconducting magnets placed over 100 meters under the ground and a no less sophisticated shielding system aiming to reduce unwanted outside influences.

Many experiments in *hard sciences* do not require randomization, as the objects the experimenters deal with are considered to be governed by universal principles and the observed effects are often assumed to be generalizable to other conditions. For instance, if a properly designed experiment concerning some universal principles is conducted in Paris, the results are assumed to be replicable in Kyiv, New York, and Shanghai.

Things are different in medicine, social sciences, and agriculture. The objects studied in these fields are usually complex and might significantly differ from each other along dimensions that are not observed. If the treatment and control groups differ systematically along some of these dimensions, this might lead to biased results. How to address this risk?

Randomized controlled trials (RCTs)

Randomized controlled trials (RCTs) are experiments where participants (or – more broadly – units) are randomly assigned to experimental conditions. The idea of randomization comes from the iconic British polymath Sir Ronald Aymler Fisher. Fisher used the idea of randomization in the context of agriculture. Although the idea was known before, he formalized it and popularized it through his work, starting in the 1920s.

The goal of randomization is to eliminate bias and permit valid significance testing (Hall, 2007). Contrary to popular misconception, randomization *does not* make treatment and control groups *balanced* or *homogenous* (Senn, 2020).

RCTs are often considered the *golden standard* for inferring causal relationships.

That said, the idea of randomization seems controversial to many. As expressed by Stephen Senn, “(...) *it has remained controversial ever since it was introduced by R.A. Fisher into the design of agricultural studies in the 1920s and even statisticians disagree about its value*” (Senn, 2021).

One of the challenges with RCTs pointed out by some authors is that they do not provide us with estimates of the efficacy of treatment at the individual level, but only at a group level (e.g., Kostis & Dobrzynski, 2020). Other authors suggest that this is not always problematic and that often the group-level conclusions are *surprisingly transportable* to individuals (e.g., Harrell, 2023), but as we saw in *Chapter 9*, group-level conclusions can hide important information from us.

In this section, we implicitly assumed that an RCT consists of two (or more) groups that are assigned to the treatment or control conditions in parallel. This is not the only possible RCT design. Moreover, other designs such as cross-over design might be more powerful. Further discussion on experimental designs is beyond the scope of this book. Check Harrell (2023) for a starter.

From experiments to graphs

Scientific experiments can help us understand which real-world phenomena are causally related. Although from a traditional Popperian point of view, it would be difficult to argue that science can provide us with *true* causal graphs, most people would probably agree that science can *help* us decide which causal graphs are *plausible*.

Each new hypothesis and each new experiment can potentially deepen our understanding of real-world causal systems and this understanding can be encoded using causal graphs (note that not all systems can be easily described using acyclic graphs and sometimes causality might be difficult to quantify). For instance, you might be interested in how the brightness of colors on your website impacts sales. You might design an A/B test to test this hypothesis. If the results indicate an influence of color brightness on sales, you should add an edge from brightness to sales to a graph describing the causes of sales.

Simulations

Simulations are another way to obtain causal knowledge.

Simulations usually start with a set of known (or hypothesized) low-level mechanisms. We code these mechanisms in order to observe a high-level behavior of a system of interest.

For instance, at TensorCell (<https://bit.ly/TensorCell>), a research group founded by a Polish mathematician and AI researcher Paweł Gora, we used simulators of traffic systems to optimize traffic control systems using various machine learning and reinforcement learning techniques.

Performing interventions in a simulator is typically much cheaper than performing interventions in a real-world system, yet if the simulator is precise enough, we expect to obtain valuable real-world insights from such interventions.

Interestingly, complex simulations might also become too expensive for certain purposes. For example, using a complex traffic simulator to produce inputs for a reinforcement learning agent is inefficient and very slow. At TensorCell, we used **graph neural networks** and **Transformer** models to approximate the outputs of simulators to tackle this problem.

In this section, we discussed the foundations of the scientific method and core scientific tools that can be used to generate causal knowledge. We briefly discussed how hypotheses can be put to the empirical test using experiments and what the main advantages and limitations of this approach are.

Now, let's take a look at less formal sources of causal knowledge.

Personal experience and domain knowledge

We started this chapter by talking about how babies perform experiments to build causal world models. In this section, we'll look into an adult's approach to refining and building such models.

Imagine a rainy chilly afternoon somewhere in Northern Europe. You stand at a bus stop near your favorite park. There's a large puddle on the street in front of you. You notice a car approaching from the left, driving close to the sidewalk. It seems that it will drive straight into the puddle. As the car approaches the puddle without slowing down, you instinctively jump behind the bus stop's shelter. Water hits the shelter's glass right next to your face, but fortunately, you are safe on the other side.

Your reaction was likely a result of many different factors, including a fast instinctive response to a threatening stimulus (splashing muddy water), but it was likely not entirely instinctive. You noticed a car in advance and likely simulated what will happen. A simulation like this requires us to have a world model.

In this case, your model of what will happen when a relatively hard object (a tire) moves through a puddle at a sufficient speed likely comes from various different experiences. It might have experimental components (such as you jumping into puddles as a 2-year-old and observing the effects of your interventions) and observational components (you seeing a person splashed by a car and observing the consequences).

Personal experiences

Personal experiences such as the ones that led you to simulate the car splashing you with the water can be a valid source of causal knowledge. Humans and some other animals can effectively generalize experiences and knowledge from one context to another.

That said, the generalizations are not always correct. A good example comes from the realm of clinical psychology. Children growing up in dysfunctional families might (and usually do) learn specific relational patterns (how to relate to other people in various situations).

These patterns are often carried into adulthood and *transferred* to relationships with new people, causing unstable or destructive relationship dynamics. Although the original patterns worked well in the family of origin, they do not work well in a new relationship. The model has not been updated.

Personal experiences are vulnerable to many biases, especially when they rely on observational data and are not examined critically. Daniel Kahneman and Amos Tversky collected an extensive list of such biases. Here, we'll only briefly discuss one of them.

An availability heuristic is a mental shortcut (usually used unconsciously) that relies on the data that comes to mind first (or most easily; Kahneman, 2011). For instance, if you usually see men rather than women in leadership positions, the image of a man will be cognitively more accessible for you when you think about leadership.

This might make you subconsciously associate men with leadership and manifest itself in a conscious belief that being male is related to being better equipped to take leadership positions, although the evidence suggests otherwise (e.g., Zenger & Folkman, 2020).

An availability heuristic might lead not only to faulty world models but also to interesting paradoxes. Multiple studies have shown that although women are perceived as having stronger key leadership competencies than men, they are not necessarily perceived as better leaders by the same group of people (e.g., Pew Research Center, 2008).

Domain knowledge

Domain knowledge can rely on various sources. It might be based on scientific knowledge, personal experiences, cultural transmission, or a combination of all of these.

Domain experts will usually have a deep understanding of one or more areas that they spent a significant amount of time studying and/or interacting with. They might be able to accurately simulate various scenarios within their area of expertise.

The main risks related to domain expertise as a source of causal knowledge are similar to the risks that we've discussed in terms of personal experience. Cultural transmission that takes place in everyday life and organizations can also equip us with incorrect or biased models (e.g., prayers for rain seem to assume an incorrect causal model).

On top of this, experts can be sensitive to overconfidence, especially when they have a narrow focus. It turns out that experts that are specialized in a narrow field tend to give predictions about the future that are not better than the ones provided by complete laymen and sometimes even worse, indicating inaccurate world models (Tetlock, 2005; Tetlock & Gardner, 2015).

Phillip Tetlock's research that we refer to here was specifically focused on forecasting in complex real-world scenarios. Although the conclusions might not generalize perfectly to some other areas, they definitely highlight an important risk factor in using domain expertise as a source of causal knowledge.

Summarizing, personal experiences and domain expertise can be valuable sources of causal knowledge. At the same time, they are susceptible to numerous distortions associated with heuristics and biases.

Personal experiences and domain expertise seem to be generally less trustworthy than scientific insights, but in certain cases, they might be in fact more accurate (e.g., when no scientific insights are available for a given (sub)domain or the domain is highly heterogeneous).

Causal structure learning

The last source of causal knowledge that we will discuss in this chapter is causal structure learning. **Causal structure learning** (sometimes used interchangeably with **causal discovery**) is a set of methods aiming at recovering the structure of the data-generating process from the data generated by this process. Traditional causal discovery focused on recovering the causal structure from observational data only.

Some more recent methods allow for encoding expert knowledge into the graph or learning from interventional data (with known or unknown interventions).

Causal structure learning might be much cheaper and faster than running an experiment, but it often turns out to be challenging in practice.

Many causal structure learning methods require no hidden confounding – a condition difficult to guarantee in numerous real-world scenarios. Some causal discovery methods try to overcome this limitation with some success.

Another challenge is scalability – the space of possible **directed acyclic graphs (DAGs)** grows super exponentially according to the number of nodes in the graph.

An exciting and relatively new research direction is to combine causal structure learning with domain knowledge and efficient experimentation.

We'll learn more about causal discovery in the next chapter.

Wrapping it up

In this chapter, we discussed three broad sources of causal knowledge: scientific insights, personal experiences and domain knowledge, and causal structure learning.

We saw that humans start to work on building world models very early in development; yet not all world models that we build are accurate. Heuristics that we use introduce biases that can skew our models on an individual, organizational, or cultural level.

Scientific experiments are an attempt to structure the process of obtaining knowledge so that we can exclude or minimize unwanted interferences and sources of distortion.

Unfortunately, experiments are not always available and have their own limitations. Causal structure learning methods can be cheaper and faster than running experiments, but they might rely on assumptions difficult to meet in certain scenarios.

Hybrid methods that combine causal structure learning, domain expertise, and efficient experimentation are a new exciting field of research.

Let's see how to implement causal discovery algorithms and how they work in practice.

See you in *Chapter 13*!

References

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